ANALYTICAL CUMUALTIVE EXAM

March 10, 2000

In a potentiometric titration the electrical potential at an indicator electrode is used to monitor the concentration of the species in solution.

$$E = E^o - \frac{0.0591}{n}\log\frac{[R]}{[O]}$$ for the reaction: $O + ne^- \leftrightarrow R$

where O and R are monitored under equilibrium conditions at the electrode surface. During a the titration the electrical potential of the solution shifts from one dominated by the redox potential of the species found in the beaker to one dominated by the titrant.

Consider the titration of Fe^{2+} (analyte) by MnO_4^-(titrant). The overall reaction in the beaker is:

$$MnO_4^- + 5Fe^{2+} + 8\,H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

The half reactions are:

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ $\qquad E^o = 0.68$ V

$MnO_4^- + 5e^- + 8\,H^+ \rightarrow Mn^{2+} + 4H_2O$ $\qquad E^o = 1.507$ V

Suppose that you titrate 0.400 L of 0.00375 M Fe^{2+} in 1.0 M HNO_3 with 0.02 M $KMnO_4$. The potential of the solution is monitored with a Pt indicator electrode and a calomel reference electrode. The potential difference measured between the two electrodes is expressed by the equation:

$$E_{solution} = E_{indicator} - E_{calomel}$$

$E_{calomel}$ is a constant 0.241 V.

1. (10 pts) Calculate the volume of titrant needed to reach the equivalence point.

2. (30 pts) Pick the titrant volume 2 mL **BEFORE** the equivalence point volume is reached and calculate the potential of the solution, $E_{solution}$.

3. (30 pts) Calculate the potential of the solution, $E_{solution}$ **AT** the equivalence point volume.

4. (30pts) Pick the titrant volume 2 mL **AFTER** the equivalence point volume is reached and calculate the potential of the solution, $E_{solution}$.